

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q22-2151

Moshers 138 kV
April 16, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2151. The Interconnection Customer proposes a 83.4 MW battery energy storage facility interconnecting at the Moshers 138 kV substation in the Pennsylvania area. The request was received on August 05, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 543 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 16 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,725,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Keswick-Walmart 138 KV line: 149.0% loading (242 MVA) following the outage of the Medora-P.T.S 345 KV line; DFAX = 24.7%.

Arsenal Hill-Keswick 138 KV: 143.6% loading (268 MVA) following the outage of the Medora-P.T.S Substation 345 KV; DFAX = 24.7%.

Forbing Rd-Arsenal Hill Substation 138 KV: 140.1% loading (288 MVA) following the outage of the P.T.S-Medora 345 KV; DFAX = 24.7%.

The Forbing Rd-Whillock Substation 138 KV is overloaded to 136.9% of its 309 MVA emergency rating for the loss of the Medora-P.T.S 345 KV. The project contributes 24.7% DFAX (20.6 MW) to this overload.

Pulliam Sw Yd-Southside Generating Sta 138 KV: 118.9% loading (265 MVA) following the outage of the Whillock Substation-Forbing Road 138 KV Line; DFAX = 8.2%.

Shepherds Central-Pulliam Sw Yd Substation 138 KV: 116.5% loading (260 MVA) following the outage of the Forbing Rd-Whillock 138 KV; DFAX = 8.2%.

The Shepherds Central-Woodville 138 KV is overloaded to 113.1% of its 257 MVA emergency rating for the loss of the Forbing Rd-Whillock 138 KV. The project contributes 8.2% DFAX (6.9 MW) to this overload.

For the loss of Simpson Stuart Tap Substation-Kempster 138 KV, the Tuco-John Day 138 KV loads to 110.9% of its 389 MVA rating. Project DFAX: 5.7%.

The Matanzas-John Day 138 KV line is overloaded to 106.3% of its 396 MVA emergency rating for the loss of the Simpson Stuart Tap-Kempster 138 KV. The project contributes 5.7% DFAX (4.7 MW) to this overload.

Artesia-Lime Crk 138 KV: 105.6% loading (264 MVA) following the outage of the Cadillac Renewable Energy-Hampton Rd Substation 138 KV; DFAX = 20.8%.

The Hampton Rd-Cadillac Renewable Energy 138 KV is overloaded to 104.4% of its 276 MVA emergency rating for the loss of the Lime Crk-Artesia 138 KV. The project contributes 25.3% DFAX (21.1 MW) to this overload.

For the loss of Tuco-John Day 138 KV, the Kempster Substation-Simpson Stuart Tap 138 KV loads to 104.1% of its 499 MVA rating. Project DFAX: 7.1%.

Augusta Electric Plant No 1 Substation-Wineagle 138 KV: 104.1% loading (349 MVA) following the outage of the Medora Substation-P.T.S 345 KV; DFAX = 5.3%.

Boyntonville-Artesia 138 KV: 101.5% loading (260 MVA) following the outage of the Hampton Rd-Cadillac Renewable Energy 138 KV; DFAX = 20.8%.

Richardson Woodhaven-Fishkill Plains 138 KV: 100.3% loading (281 MVA) following the outage of the Simpson Stuart Tap-Fishkill Plains 345 KV Line; DFAX = 14.8%.

The Filer 500/345kV XFMR is overloaded to 100.2% of its 1,238 MVA emergency rating for the loss of the Lerdo-Tidd Substation 500 KV L. The project contributes 7.5% DFAX (6.3 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Keswick-Walmart 138 KV line	Rebuild line with higher-capacity conductor	\$1,160,000	18% (\$215,000)
Arsenal Hill-Keswick 138 KV	Rebuild line with higher-capacity conductor	\$1,744,000	18% (\$323,000)
Forbing Rd-Arsenal Hill Substation 138 KV	Rebuild line with higher-capacity conductor	\$1,516,000	18% (\$280,000)
Forbing Rd-Whillock 138 KV	Rebuild line with higher-capacity conductor	\$1,850,000	18% (\$342,000)
Pulliam Sw Yd-Southside Generating Sta 138 KV	Rebuild line with higher-capacity conductor	\$1,240,000	6% (\$78,000)
Shepherds Central-Pulliam Sw Yd 138 KV	Rebuild line with higher-capacity conductor	\$1,585,000	6% (\$99,000)
Shepherds Central-Woodville 138 KV	Rebuild line with higher-capacity conductor	\$815,000	6% (\$51,000)
Tuco-John Day 138 KV	Rebuild line with higher-capacity conductor	\$1,358,000	8% (\$111,000)
Matanzas-John Day 138 KV line	Reconductor line	\$583,000	8% (\$48,000)
Artesia-Lime Crk 138 KV	Reconductor line	\$523,000	12% (\$65,000)
Hampton Rd-Cadillac Renewable Energy 138 KV	Reconductor line	\$1,040,000	16% (\$171,000)
Kempster Substation-Simpson Stuart Tap 138 KV	Reconductor line	\$2,431,000	13% (\$327,000)
Augusta Electric Plant No 1-Wineagle 138 KV Line	Reconductor line	\$1,637,000	14% (\$232,000)
Boyntonville-Artesia 138 KV	Reconductor line	\$1,038,000	12% (\$129,000)
Richardson Woodhaven-Fishkill Plains 138 KV	Reconductor line	\$932,000	32% (\$294,000)
Filer 500 kV/345 kV XFMR	Replace transformer with higher-rated unit	\$35,137,000	3% (\$960,000)
	Total allocated to project		\$3,725,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.